

Technical University of Denmark
Department of Applied Mathematics and Computer Science

02170 Database Systems
Practice Examination Part 1 — Exam A

Aids allowed: All
Duration: 3 hours
Weighting: Part 1: 50%, Part 2: 50%

Note: unless explicitly asked for, SQL queries should not return tables having duplicate rows.

A database for a Library Network

A database named **LibraryDB** contains information about libraries, members, books, and loans.

A **Library** has a unique name (*lname*) and is located in a city (*city*). A **Member** has a unique id (*mid*), a name (*mname*), and is registered at a single library (identified by *lname*). A **Book** has a unique id (*bid*), a title (*btitle*), and a publication year (*byear*).

The **borrow**s relation records which members have borrowed which books. The **review**s relation records which members have reviewed which books. Note that a member can borrow or review books from any library.

Relation schemas:

Library(lname, city)

Book(bid, btitle, byear)

Club(cname, lname)

Member(mid, mname, lname)

borrow(s(mid, bid)

attends(cname, mid)

Library

lname	city
CentralLib	Copenhagen
NorthLib	Copenhagen
AarhusLib	Aarhus
OsloLib	Oslo

Member

mid	mname	lname
1	Anna	CentralLib
2	Bo	CentralLib
3	Carla	AarhusLib
4	Daniel	NorthLib
5	Eva	OsloLib

Book

bid	bttitle	byear
101	Database Design	2020
102	Algorithms	2018
103	Clean Code	2019
104	AI Fundamentals	2022

borrow(s

mid	bid
1	101
1	102
2	101
3	101
3	103
4	102

Club

cname	lname
ReadersCircle	CentralLib
BookNerds	AarhusLib
PageTurners	NorthLib

attends

cname	mid
ReadersCircle	1
ReadersCircle	2
BookNerds	3
PageTurners	4

Questions

Question 1

Write an SQL query that returns a table containing, for each member who is registered at a **Copenhagen** library and who has **never borrowed any book**, his/her **mname** and **lname**.

Question 2

Write an SQL query that returns a table containing, for each book that has been borrowed by **2 or more members**, the **bid**, **btitle**, **byear**, and the total number of members who borrowed it. Name the count **numBorrowers**.

Question 3

Write an SQL query that returns a table containing, for each club, the **cname** and the number of **distinct books borrowed by at least one member attending that club**. Name the count **numBooks**.

Important: every club must appear in the result, even those whose members have not borrowed any book.

Question 4

Create an SQL function named **bookCount**, which takes the **mid** (of type **INT**) of a member as argument and returns the number of books borrowed by that member. If the member does not exist, it should return 0. The returned type should be **INT**.

*Do not include any **DROP FUNCTION** or **SELECT** statements using the function.*

Answers Template (answers_A.sql)

```
-- Student ID:  
USE LibraryDB;  
  
-- q1:  
-- q2:  
-- q3:  
-- q4:
```